State fixed effects

Controls: Use State Fixed Effects; Fixed Effects Type — Modern or Historical.

1. Fixed effects change the comparison behind the estimate. For a modern county or MSA g, let y_g be the selected log GDP per employment outcome, including any sector and human-capital choices. Write $d_g = N_g / A_g$ for density, using total employment N_g and land area A_g. A regression with state fixed effects can be written as

$$y_g = \alpha_{s(g)} + \rho \log d_g + W_g' \delta + u_g.$$

Here $\alpha_{s(g)}$ is a separate intercept for the state group assigned to g, W_g contains the selected controls, and ρ is the density elasticity. The estimate then uses variation between observations within the same state group after accounting for controls. In IV, the historical instrument must also explain useful variation within those groups.

2. Modern and historical effects define different groups. *Modern* uses current state identities. *Historical* uses the historical state or territory in the selected territory year. Changing only the instrument year leaves these historical groups unchanged. For counties, the historical assignment locates the county's geographic centre in the historical jurisdictions, with an interior-point fallback when the centre lies outside all mapped jurisdictions.

An MSA crossing several states receives one group, rather than several fixed effects. The project's defining county unit is the member with the highest 2010 employment density. Its state determines the modern group, and its geographic centre determines the historical group. Thus, MSA fixed effects do not separately remove every constituent state's contribution.

3. Fixed effects can reduce precision and the sample. State effects absorb differences shared within a group. If most historical instrument variation was between states, less remains to identify density after these effects are included. The coefficient or standard error can therefore change substantially. Observations without a usable required identity are excluded. Ordinary linear regressions also remove singleton fixed-effect groups, which contain only one usable observation and cannot contribute within-group variation.

4. State-level CH and county calibration differ. There is one observation per modern state in a selected year. A separate fixed effect for every state would absorb all state-level variation, so the website does not offer it. Historical state or territory group effects are also not implemented for State CH. County-adjusted CH nevertheless uses modern state fixed effects in the linked county calibration, where there are multiple counties per state. These county-calibration intercepts are not inserted into the regional CH index.